

Path to final product

Milestone 1

When the assignment brief of Milestone 1 was first released, the first thing we did was organize weekly Zoom meetings on Thursday morning. In the first meeting, we gave ourselves one week to come up with three datasets each for a total of nine datasets, which we would then shortlist to one dataset to work with. The list below was the list of datasets we came up with:

- Water flow in Swiss rivers and lakes
- Zurich weather station
- Pokémon
- World population
- Exoplanets
- OECD better life index
- Titanic
- Student dropout and academic success
- Obesity levels

We shortlisted it to the weather, OECD and obesity datasets using a vote. With another vote, it was then decided to do the OECD better life index dataset. A GitHub repository was then set up, and we evenly split the work for Milestone 1 report as follows:

- Dataset section: Francesco
- Problematic section: All
- Exploratory Data Analysis section: Corvin
- Related Work section: Nicholas

It is to be noted that during the EDA section, Corvin cleaned the dataset and generated the histograms needed for the Milestone 1 report. We finished Milestone 1 on the 13th of March 2025, or about one week before the deadline.

Milestone 2

On the first week of Milestone 2, we first discussed what the core visualization MVP would look like, as well as possible extra ideas that would elevate the visualizations. We ended up with the core design features of the MVP as follows:

- a 2D map of the whole world with all countries
- sliders – one for the year, and 11 more for the individual OECD metrics
- a color scale that would indicate how good or bad the countries scored on the chosen metric(s)
- countries with no data for the chosen metric(s) would be grayed out
- show tooltip of basic information when hovering over a country
- show some visualizations using the metrics when clicked on a country

and the possible additional features in the future:

- a 3D map of the whole world
- an extended view for subcategories
- animations and transitions for a smoother user experience

That same meeting, we decided to do the first section with the two A4 pages, and to postpone dealing with the prototype until after the Milestone 2 report was complete. Division of labor for the report was as follows:

- Sketches: Corvin
- Listing of tools needed: Francesco
- Break down goal into independent pieces to implement: Nicholas

However, over the weekend, Francesco built and committed a website skeleton to the repository without prior consultation with Corvin or Nicholas, while also maintaining his commitment in listing the tools needed. The website was also hosted on GitHub. This prototype included the 2D map of the whole world with the countries, text containing name of country and score in metric chosen when hovering over a country, visualizations when clicked on a country in the form of a metric-time line graph and a radar chart, and a time slider for the years. The prototype Francesco created also had a slight modification – instead of 11 individual sliders for each OECD metric, there was a single unified drop-down menu for users to select the OECD metric for the visualization to display. This change was done to make the project visualization more distinct from the flower visualization already on the official OECD website.

Due to this surprise jump in progress, in the next weekly meeting, it was decided that Corvin would handle the dataset preprocessing and backend dataset integration into the skeleton, while Nicholas would write the entire report for Milestone 2. For Corvin, this also involved normalizing all the metrics, which used this formula he came up with:

```
def _normalise(group: pd.DataFrame) -> pd.Series:
    v = group["OBS_VALUE"].astype(float)
    lo, hi = v.min(), v.max()
    if np.isclose(lo, hi):
        return pd.Series(0.5, index=group.index, dtype=float)
    scaled = (v - lo) / (hi - lo)
    if group["Measure"].iat[0] in NEGATIVE_MEASURES:
        scaled = 1 - scaled
    return scaled
```

The core features missing from the Milestone 2 skeleton at this point were the graying out of countries for no data and the color scale for the metric rankings. By his own initiative, Corvin managed to implement the color scale for the site for Milestone 2, but there were still some CSS bugs encountered, so it did not work perfectly. These bugs would later be ironed out by Francesco during Milestone 3.

Milestone 3

Before the first weekly meeting of Milestone 3, Francesco unilaterally did some refactoring and bug fixes to the code, as well as added some new features: dark mode, color-blind color scale, and a Play button to automatically go through all the years. This happened during Easter break.

During the first weekly meeting of Milestone 3, the group discussed suggestions, criticisms and potential improvements for the current iteration of the website, as follows:

- Nicholas to advise on any accessibility violations and recommended fixes
- Color-blind toggle option for users to view the visualization
- Lighter text font for dark mode
- Make information box more transparent
- Make blue square around the country when selected disappear faster, as a bug where it lingers was found
- Use OpenDyslexic font
- Add The Czech Republic and Colombia back into the dataset, as there was an undetected naming convention bug
- Add 3D globe option

Division of labor for that week follows:

- Process book: Nicholas
- SVG icons to add into the drop-down menu: Corvin
- Fixing dataset naming convention bug: Corvin
- Blue square highlight bugfix: Francesco
- Color-blind toggle: Francesco
- Lighter text font for dark mode: Francesco

Corvin later sent the site for OpenDyslexic font to a dyslexic friend of his, and this friend mentioned that he found no difference between OpenDyslexic and standard sans-serif fonts. As such, the idea to use OpenDyslexic was dropped. Feedback for Milestone 2 was also received a few days after the first weekly meeting. Since the feedback recommended us to work on our overall storytelling, we decided to focus on this and dropped the 3D globe idea.

“You have to work on the message. You could for example show interesting facts or extremal behavior on your dataset. You could also make a comparison of two countries.”

Based on this feedback, we also decided that it would be a good idea to include a landing page with the storytelling elements inside it. From this point onwards, the group members started acting unilaterally and independently to improve the site. The previously agreed commitments from the first week’s meeting were still valid and were all met. But due to everyone having multiple competing deadlines from other courses in May, it was implicitly decided that whoever could work towards improving the site would do so.

Francesco added a country comparison feature for the main visualization page, created the entirety of the landing page and introduced transitions and animations to both webpages. Corvin ran into challenges integrating the SVG icons into the drop-down menu, and thus had to re-write the entire drop-down menu code to create a custom drop-down menu that was compatible in displaying his created SVG icons. Nicholas noted accessibility violations in both the visualization page and landing page. Nicholas would then go back and forth with Corvin and Francesco to decide which accessibility hotfixes to implement to the landing page. Nicholas also improved some of the icons in the landing page.

By early May, the team was unsure whether the current feature-set met grading expectations, so we attended the TA meeting on the 9th of May as a group and consulted with the professor and TA. Here we found out that our site was mostly good enough, just lacking a small feature where extrema top and bottom 5 countries for the selected metric for that year should be displayed in the visualization page. It was confirmed in this meeting that accessibility can largely be ignored for the scope of the project and still receive a good grade. Nevertheless, Nicholas went ahead with the accessibility hotfixes for the landing page, since he had already implemented half of the agreed accessibility feature hotfixes. However, planned accessibility fixes to the visualization page were dropped because of the information from this meeting about how accessibility affected grading. Below is the list of accessibility features Nicholas successfully implemented to the landing page:

- Added missing alt text for all svg icons
- Removed redundant aria-label from sun and moon icons for theme toggle
- Changed font colors of blue non-hyperlink text
- Removed italic font for dyslexic users
- Changed some font styles to replace italic to highlight certain sections of text
- Made landing page fully navigable with keyboard only
- Made landing page fully compatible with high zoom levels, up to 500%
- Made landing page fully compatible with mobile devices

Nicholas also pointed out two major bugs in the visualization page. If the user inputs mouse scroll wheel inputs while the transition of clicking off a country was still playing, it would brick the webpage. The user can either click on that same country again or reload the entire webpage. The transition animation for the line graph and radar chart would always play when using the time slider. Corvin also pointed out a limitation that the dashboard with the line graph and radar chart were hardcoded in JavaScript, which meant the site could break with devices with larger viewports, just like his own monitor. These were all unintended, so Francesco fixed the transitions. However, he had no clue what was causing scroll wheel inputs to brick the site. And at this point, the CSS has gotten so out of hand due to rapid and independent feature additions and lack of a unified styling strategy from the outset that the rest of these bugs were deemed not worth fixing by the group.

Francesco implemented the extrema top and bottom 5 country feature into the visualization page, and added GitHub links to both the landing page and visualization page. Nicholas applied one last hotfix to make the GitHub links not open a new tab/window, added title tags to the GitHub links, and made the extrema feature open by default. At this point, the state of the site was considered final. The group then worked on the screencast and thus concluded our Milestone 3.

Challenges Faced

One of the most persistent challenges in this project was managing the CSS. As the codebase grew, the styles became increasingly complex and interdependent, making it difficult to implement design changes without unintentionally affecting other elements; especially when introducing and styling new functionality. Existing CSS bugs compounded the issue, and a significant amount of time was spent debugging and refactoring styles to achieve a consistent and functional layout. This considerably slowed down the development process and affected our overall pace. This experience highlighted the need for a more structured CSS methodology like BEM, utility-first CSS, or CSS-in-JS in future projects to manage scalability.

For quite some time, there was an undetected discrepancy between the naming convention of certain countries from the original dataset to the actual names of these countries. For example, **The Czech Republic** was named **Czechia** in the dataset. This caused The Czech Republic not to appear on the Milestone 2 skeleton and the WIP Milestone 3 website. The bug was eventually discovered during a weekly meeting when Corvin pointed out that certain countries that were in the dataset were not appearing on the map, prompting an immediate dataset review and correction of the naming mismatches. This was only possible due to Corvin's familiarity with the dataset and his attention to detail.

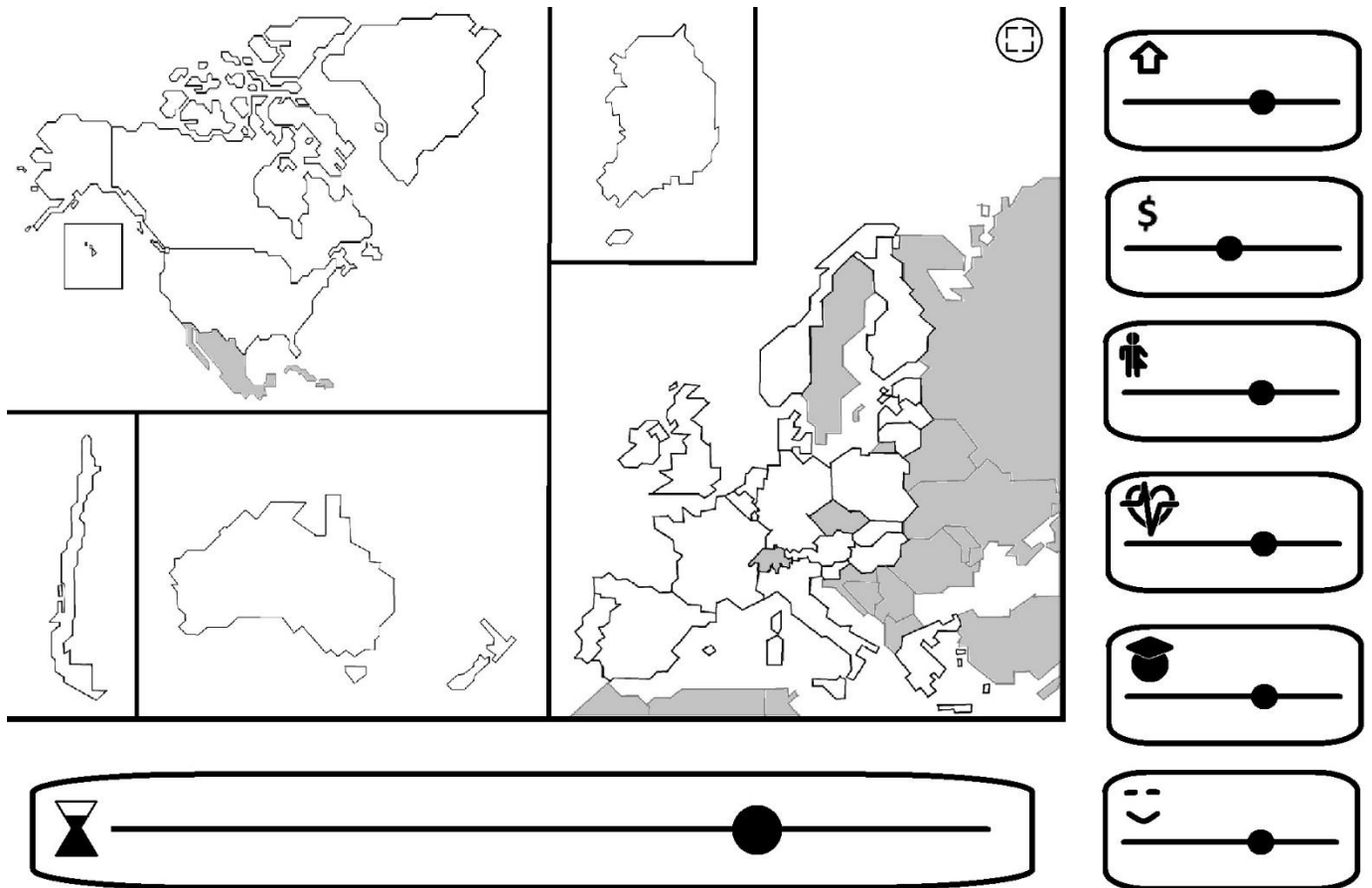
Our initial idea to include a 3D globe visualization ultimately proved unfeasible. As early as Milestone 2, it became clear that implementing this feature would be technically complex and time-consuming. As such, we designated this as an "extra idea" that could maybe be implemented sometime in the future. However, by Milestone 3, competing deadlines from other courses, combined with the technical demands of the feature and Milestone 2 feedback emphasizing the need for stronger narrative elements, led us to officially drop this idea from the project scope.

Accessibility was also a major challenge in this project. Due to time constraints from competing course deadlines and the very broad scope of the Web Content Accessibility Guidelines (WCAG) 2.2, we were unable to achieve full compliance and decided to prioritize core technical functionality instead. With Francesco and Corvin less familiar with web accessibility practices, Nicholas took responsibility for iteratively reviewing the site and identifying accessibility violations. These violations would then be discussed as a group to determine which issues could realistically be addressed within the scope of the project. These approved hotfixes were then implemented by Nicholas to improve the overall accessibility of the site.

Notable accessibility limitations that remained out of scope include:

- lack of mobile optimization for visualization webpage
- absence of testing with screen readers or speech recognition tools
- absence of testing for compatibility with accessibility color filters and high contrast modes
- absence of testing with OS-level accessibility settings such as text size
- absent compatibility for keyboard users to navigate the map visualization
- absent compatibility of using the zoom functionality at all with the visualization country charts, since those were hardcoded

Sketches



Our initial sketches from Milestone 2 accurately guided the development of the skeleton and design of the visualization page in the final product. We found our early planning was robust, and no significant deviations from these visual plans were necessary for the core functionality. The primary evolution occurred in the technical implementation and data handling rather than the fundamental layout depicted in the sketches, especially since the sketch provided was a combination of all the different dynamic features into a single sketch. The only notable difference here would be the use of a drop-down menu for the individual metrics instead of sliders.

Design of process book

The creation of this process book was not merely about documenting our project journey, but also about presenting that journey in a clear, professional, and accessible manner. The design decisions for this document were consciously made to enhance readability and ensure a consistent visual experience, primarily utilizing the capabilities of Microsoft Word.

We began with the selection of Microsoft Word as the primary authoring tool, due to its widespread availability and robust formatting options suitable for structured documents. We opted for a standard A4 page size combined with Narrow margins (half inch on all four corners). This configuration maximizes the usable content area on each page, allowing for more information to be presented clearly and accommodating wider sketches or diagrams. A blank white background was maintained throughout, ensuring content legibility and a clean, professional aesthetic without distracting visual elements.

The consistent application of font formatting was crucial for establishing visual order and guiding the reader. For headings, 26pt Calibri centered, bold and underlined was used for section titles. 16pt bold Arial left-aligned was used for subheadings. For all main body text, 12pt Arial left-aligned was selected due to its standard use in documents.

Careful attention was paid to spacing to enhance the document's flow and prevent text from appearing dense. Body text paragraphs were formatted with Single line spacing with 12pt spacing After, which provided sufficient white space between lines for comfortable reading and created clear visual separation between distinct blocks of text. We also strategically used bullet points to break down complex information into digestible segments and highlight key features and decisions.

Peer Assessment

- Sketches: Corvin
- Site feedback and suggestions: Corvin, Francesco, Nicholas
- Accessibility advisor: Nicholas
- Testing/QA: Corvin, Nicholas
- Milestone 2 report: Nicholas
- Process book: Nicholas
- Milestone 2 website skeleton
 - 2D map of the whole world with the countries: Francesco
 - Information display when hovering over a country: Francesco
 - Line graph and a radar chart: Francesco
 - Color-scale for comparing country performance: Corvin, Francesco
 - Time slider: Francesco
- Dark mode toggle for all webpages: Francesco
- Color-blind color-scale: Francesco
- Play button for time slider: Francesco
- Landing page skeleton:
 - Header: Francesco, Nicholas
 - Icon design: Francesco, Nicholas
 - Content of site: Francesco
- Landing page fixes:
 - Alt text for all svg icons: Nicholas
 - Removed redundant aria-label from sun and moon icons: Nicholas
 - Changed font colors of blue non-hyperlink text: Nicholas
 - Removed italic font for dyslexic users: Nicholas
 - Changed some font styles to replace italic for highlighting: Nicholas
 - Made landing page fully navigable with keyboard only: Nicholas
 - Made landing page fully compatible with high zoom levels, up to 500%: Nicholas
 - Made landing page fully compatible with mobile devices: Nicholas
 - GitHub hyperlink title tags: Nicholas
 - GitHub hyperlink not opening new tab/window: Nicholas
- Dataset processing: Corvin
- Dataset integration: Corvin
- SVG image design and implementation: Corvin
- Implementation of custom drop-down list for SVG compatibility: Corvin
- Webpage animations and transitions: Francesco
- Extrema Top/Bottom5 display: Francesco
- Visualization page fixes:
 - Transitions with time sliders no longer applied: Francesco
 - GitHub hyperlink title tags: Nicholas
 - GitHub hyperlink not opening new tab/window: Nicholas
 - Extrema display open by default: Nicholas